Wave speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use speed, frequency and wavelength in a consistent relationship.

SUCCESS CRITERIA - I can...

1. Use v=fλ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A system absorbs 70 J heat and does 30 J work. Find its internal-energy change using $\Delta U = Q - W_{by}$.

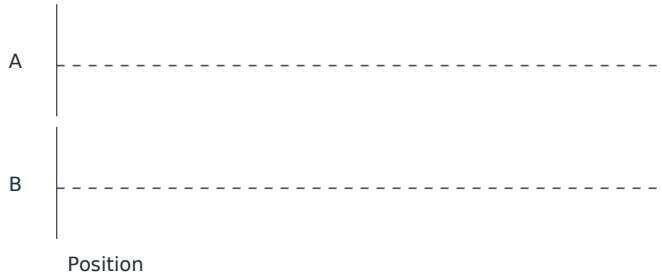
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: v = f λ; λ = v/f.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

Worked reasoning: A wave with wavelength 3 m and frequency 2 Hz travels at v=f λ=6 m/s. At the same speed, frequency 3 Hz would require wavelength 2 m.

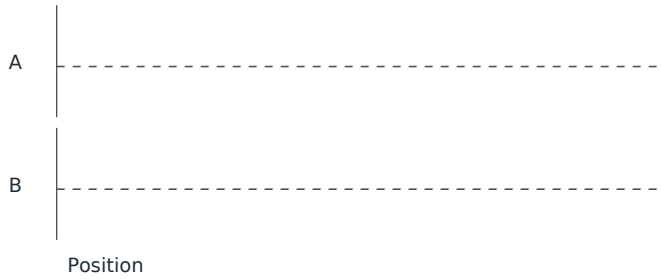
Your turn: Draw both waves over the same 6 m length and label their frequencies.



Marzano check - circle: ① ② ③

03 Read and interpret the model

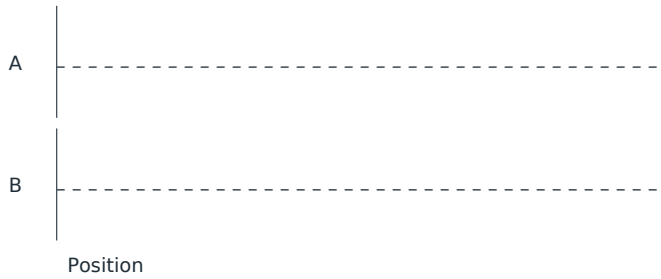
Two waves travel in the same ideal medium at the same speed. One has twice the frequency. Compare wavelengths.



Marzano check - circle: ① ② ③

04 Apply the physics

A wave simulator shows frequency 4 Hz and wavelength 2 m. Calculate wave speed. Set the frequency to 8 Hz while keeping speed fixed. Find the new wavelength and sketch both patterns over the same distance.



Marzano check - circle: ① ② ③

05 Explain your reasoning

A student writes v=f/λ. Check units and explain why that is not a speed.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Sunlight carries energy across 4 million km of nearly empty space in a simplified example. Identify the transfer mechanism and explain why matter is unnecessary. Label temperatures, samples and relevant energy transfers.

Sample A

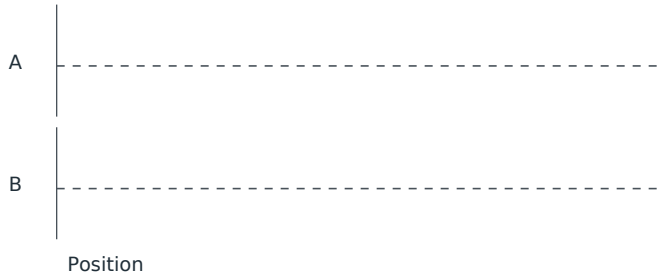
Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

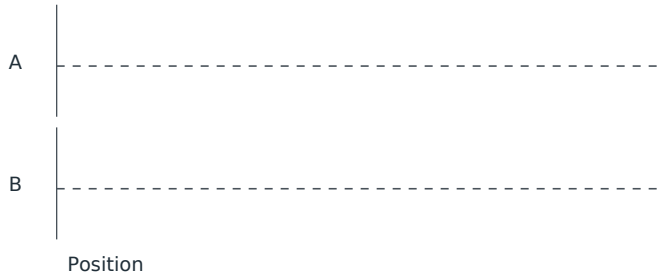
f=3 Hz,λ=5 m. Find v.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Create three frequency-wavelength pairs that all give 12 m/s. Sketch one pair.



Marzano check - circle: 1 2 3